Michaela K. Shallue

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Education

PhD, Civil and Environmental Engineering, *Utah State University* Jan 2024 – Present
River mechanics and modeling | Advisor – Colin B. Phillips
Dissertation: Disturbance-driven sediment dynamics and channel response (working title)

MSc, Geology, *Utah State University* Aug 2021 – Dec 2023
Geoscience education | Advisor – Joel Pederson
Thesis: Building and Using a Hydrology Experiment for Place-Based Learning with Native American Students
Minor: Climate Adaptation Science

BSc, Geology, Honors, *Northern Arizona University* Aug 2015 – May 2019
Fluvial geomorphology | Advisor – Taylor Joyal
Thesis: Hydrologic impacts of climate and land use in Canyon de Chelly, Arizona
Minors: Honors Program, Psychological Sciences

Academic Appointments

Research and Teaching Assistant, *Utah State University* 2024 – Present
Utah Water Research Laboratory, Civil and Environmental Engineering Department

Research and Teaching Assistant, *Utah State University* 2021 – 2023
Department of Geosciences

Climate Adaptation Science Fellow, *Utah State University* 2021 – 2023
Ecology Center

Teaching Experience

Utah State University
Role: Graduate student | 5 courses

Environmental and Hydrologic Data Analysis and Experimentation, Spring 2026
CEE5930/6660. Teaching Assistant.
Course size: 15 graduate students, 1 undergraduate.

Fluid Mechanics, CEE3500. Teaching Assistant. Fall 2025
Course size: 69 undergraduates.

Geomorphology Lab, GEO/WATS 3600L. Instructor (2 years). Fall 2022, 2023
Course size: 24 undergraduates.

Geomorphology Lecture, GEO/WATS 3600. Teaching Assistant (2 years). Fall 2022, 2023
Course size: 24 undergraduates.

Field Camp: Silver Islands, UT, Project, GEO 5200/5210. Teaching Assistant. Summer 2023
Course size: 13 undergraduates (8 Utah State, 5 Weber State).

Guest lectures
Physical Geology (Fall 2025), GEO1110. Guest lecture on fluvial geomorphology.
Introduction to Geology (Summer 2022, 2023), GEO1010. Guest lecture on runoff processes.

Publications

Peer-Reviewed (1)

1. Masteller, C.C., Phillips, C.B., Kostynick, R.P., Castejon-Villalobos, J.F., Lopez, C.G., **Shallue, M.K.**, Bower, J., Sigman, A. (2026). Tracking the trajectory of alluvial channel adjustment reveals along-river shifts in sediment mobility. *Geophysical Research Letters*, 53, doi.org/10.1029/2026GL122483.

In Preparation (1)

1. **Shallue, M.K.**, Phillips, C.B. (in prep). Suspended sediment patterns reveal dynamic transport response to dam removal on the Elwha River, Washington. *Target journal: Geophysical Research Letters*.

Non-Peer-Reviewed (2)

1. Pederson, J., Tanski, N., **Shallue, M.** (2024). Tracking the timing and dynamics of Colorado River incision through study of river terraces near Dewey, UT. *Final Report on Canyonlands Natural History Association Discovery Pool Grant*.
2. **Shallue, M.** (2023). Building and Using a Hydrology Experiment for Place-Based Learning with Native American Students. *Utah State University MSc Thesis*, doi.org/10.26076/4adb-06ab.

Other (1)

1. **Shallue, M.** (2021). Earth Notes: The Returning Rapids of Lower Cataract Canyon. *Earth Notes: KNAU News and Talk – Arizona Public Radio*.

Conference Presentations

Presentations – Total: 12* | Oral: 7 | Poster: 6
*One presentation combined both formats and is counted in both categories

2026 (2)

Shallue, M., Patterson, N., Scamardo, J., Phillips, C.B. (2026). Rethinking Flood Frequency in Dryland Streams: Hydrologic and Geomorphic Drivers of Flood Hazard. 2026 Fall Meeting, AGU, *oral presentation* (accepted).

Shallue, M., Patterson, N., Scamardo, J., Phillips, C.B. (2026). Hydrologic and Geomorphic Controls on Flooding in Dryland Riverscapes. Geological Society of America Connects 2026, *poster*.

2025 (4)

Shallue, M., Scamardo, J., Phillips, C.B. (2025). Beyond the Annual Maximum Series: Hydrological and Geomorphic Flood Risk in Dryland Streams. 2025 Fall Meeting, AGU, *poster*.

Shallue, M., Phillips, C.B. (2025). Patterns in Suspended Sediment Reveal Dynamic River Transport Response to Dam Removal. 2025 Fall Meeting, AGU, *GeoBurst (oral + poster presentations)**.

Shallue, M., Pederson, J., Searle, K. (2025). Cultivating Interest in Geoscience Among Native Students with an Outdoor Hydrology Experiment. Geological Society of America 2025 Rocky Mountain Section Meeting, *oral presentation*.

Shallue, M., Phillips, C.B. (2025). Dam Removal Reveals Influence of Sediment Availability on Transport Dynamics. 2025 USU Spring Runoff Conference, *poster*.

2024 (2) **Shallue, M.**, Phillips, C.B. (2024). Dam Removal Reveals Influence of Sediment Availability on Transport Dynamics. 2024 Fall Meeting, AGU, *poster*.

Shallue, M. (2024). Dam Removal Influence on Sediment Transport. TAPDINTO-STEM Mountain Hub Convening, *oral presentation*.

2023 (1) **Shallue, M.**, Chavez, E. (2023). Evapotranspiration and Water Trends in the San Luis Valley. State of the Basin 2023, *oral presentation*.

2022 (1) **Shallue, M.**, Huntley, N. (2022). Interdisciplinary Specialization for Graduate Students Who Want to Put Science into Action: Climate Adaptation Science (CAS). Geological Society of America Connects 2022, *oral presentation*.

2019 (2) **Shallue, M.**, Joyal, T.J., Shaw, N. (2019). Canyon de Chelly: Watershed Modeling to Predict the Hydrological Impacts of Climate and Land Use. Student Water Symposium, Northern Arizona University, *oral presentation*.

Shallue, M., Joyal, T.J., Shaw, N. (2019). Canyon de Chelly: Watershed Modeling to Predict the Hydrological Impacts of Climate and Land Use. Undergraduate Research Symposium, Northern Arizona University, *poster*.

Grants

Awarded (4) – To date: $7,000 awarded as lead PI.

Utah State University | Role: Graduate Student | 3 grants (3 PI)
Geological Society of America, Graduate Research Student Grant 2026
[Student, $3,000]
Geological Society of America, Graduate Research Student Grant 2022
[Student, $2,500]
Four Corners Geological Society, Master's Degree Thesis Grant 2022
[Student, $1,000]

Northern Arizona University | Role: Undergraduate Student | 1 grant (1 PI)
School of Earth and Sustainability, Virginia L. Korte Undergraduate Research Award 2018
[Student, $500]

Student Travel Grants (4) – To date: $1,650 awarded (2024 – 2026)
Awarded from: USU School of Graduate Studies, USU College of Engineering, and the Rocky Mountain Section of the Geological Society of America, to attend the AGU Fall Meeting or GSA Connects.

Honors & Awards

Farouk El-Baz Student Research Award, Geological Society of America 2026
TAPDINTO-STEM Mentorship, TAPDINTO-STEM Alliance 2023 – 2024, 2024 – 2025
Outstanding Graduate Teaching Assistant, USU Department of Geosciences 2024
J. Stewart Williams Graduate Fellowship, USU Department of Geosciences 2022, 2023
Climate Adaptation Science Fellowship, Utah State University 2021 – 2023
Outstanding Field Camp Student, NAU Geology Department 2019
1st Place Undergraduate Presentation, NAU Student Water Symposium 2019
Mary Crawley Scholarship, NAU Cline Library 2019
Lumberjack Scholars Award, Northern Arizona University 2015 – 2019

Leadership & Service

Departmental and Organizational Service

AGU Earth and Planetary Surface Processes
- *Mentor*, AGU EPSP Mentoring Program (mentoring 3 students) — Oct 2026 – Jan 2027

Civil and Environmental Engineering Department, *Utah State University*
- *Lead Student Ambassador*, Utah Water Research Laboratory — Fall 2025 – Summer 2026
- *Student Research Symposia Judge*, USU Office of Research — Apr 2026
- *Student Ambassador*, Utah Water Research Laboratory — Fall 2024 – Summer 2025

Department of Geosciences, *Utah State University*
- *Rock and Fossil Day Volunteer* — Feb 2024, 2025
- *Project Co-Lead*, Native American Summer Mentorship Program — Summer 2022, 2023
- *URGE Pod Member*, USU URGE Refinement Committee — Fall 2022 – Summer 2023
- *Committee Member*, Ecology Center Seminar Series — Spring 2022 – Spring 2023
- *DEI Graduate Representative*, Department of Geosciences — Fall 2021 – Fall 2023
  - *Co-Lead*, Geosciences DEI Brown Bag Lunches — Spring 2023 – Fall 2023

School of Earth and Sustainability, *Northern Arizona University*
- *Research Volunteer* (post-bacc) — Sept 2019 – May 2020
- *Undergraduate Student Researcher* — Aug 2018 – May 2019

Cline Library, *Northern Arizona University*
- *Department Representative*, Library Spaces Committee — Oct 2019 – June 2021
- *Student Ambassador* — Aug 2018 – May 2019

Scientific Meetings

Geological Society of America
- *Student Volunteer*, Annual Connects Meeting — 2022
- *Student Volunteer*, Joint Section Meeting — 2018, 2022

Training

- CLaSH Summer Field School (Graduate Technical Short Course), *CLaSH* — July 2026
- EPSP Graduate Student Mentorship Program (mentee), *AGU EPSP Section* — Fall 2025
- TEACH Cultural Competence Training, *Utah State University* — Jan 2025
- Empowering Teaching Excellence (ETE), *Utah State University*
  - *ETE Explore College Teaching Program* — Jan 2024 - Present
  - *ETE Learning circles:*
    - Crucial Conversations — Oct 2026
    - The Norton Guide to Equity-Minded Teaching — Feb 2025
    - "Whatever It Is, I'm Against It": Resistance to Change in Higher Education — Oct 2024
- TAPDINTO-STEM Mentorship Program (mentee), *USU, NSF* — 2023 - 2025
- Science, Management, & Policy Exchange, *USU Climate Adaptation Science* — Mar 2022, 2023
- Glacial Geomorphology and Volcanology Field Camp, *South Dakota Mines* — July/Aug 2018

Other Service and Roles

Prehistoric Museum, *Utah State University*
- *Education and Paleontology Intern* — July 2022 – Aug 2022
- *Field and Preparatory Lab Volunteer* — Aug 2021 - Present

Science Moab
- *Media Outreach Volunteer* — Oct 2020 – Aug 2021

Sierra Club, Grand Canyon Chapter
- *Student Volunteer* — Oct 2020 – Sept 2021

Hopi Water Resources
Water Data Manager Oct 2018 – May 2019
Cline Library, *Northern Arizona University*
E-Resources Library Coordinator Dec 2018 – June 2021
Assembly Hall Student Assistant Aug 2018 – Dec 2018
Course Reserves Student Assistant May 2018 – Dec 2018
User Experience Student Assistant Nov 2015 – Dec 2018

Skills

Field Experience: Locations and Skills

Pingree Park, CO. River surveying, UAV and handheld photogrammetry 2026
Bear River Mountains, UT. River surveying, grain size, ADCP flow 2025
Eel River, CA. Grain size 2024
Dewey, UT. Optically stimulated luminescence (OSL) sampling and terrace mapping 2022, 2023
Glen Canyon, UT. OSL sampling 2022
Book Cliffs, UT. Clast counts, vegetation surveys, rainfall simulation (infiltration, runoff) 2022
Coconino National Forest, AZ. Post-fire debris flow monitoring 2021
Fort Valley, AZ. Surface water sampling 2018 - 2019
Canyon de Chelly, AZ. River surveying, stream and rain gage installation, infiltration 2018 - 2020
Other field skills: Bedrock mapping, differential GPS surveying, stadia rod surveying
Paleontological field experiences: 2021 - 2023
Fossil prospecting, quarrying, quarry mapping, and field jacketing with the Prehistoric Museum in the Book Cliffs near Green River, UT, Wasatch Plateau, UT, Big Water, UT, and Thompson Springs, UT. Associated lab experiences include vertebrate fossil preparation and archival vertebrate fossil jacketing.

Laboratory Experience

OSL sample preparation; wet and dry sieving

Software Experience

ArcGIS Pro; QGIS; Python; Adobe Creative Suite (Illustrator, Photoshop, and InDesign); GRASS GIS; MATLAB.

Memberships

American Geophysical Union
Association for Women Geoscientists
Geological Society of America
National Association of Geoscience Teachers
Society for Sedimentary Geology
Utah Geological Association

– Updated October 16, 2026 –